

Observation versus intervention

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1 Overview

Seeing that the vegetation is dense and forcing the vegetation to be dense are different operations with different consequences ([Pearl 2009](#)). `observe` records evidence and leaves the network untouched; `do_intervention` rewrites the network, replacing the mechanism of the intervened variable by a constant – the “cut” that string-diagram surgery performs on a causal model ([Jacobs et al. 2019](#); [Lorenz and Tull 2023](#)). This vignette compares the two numerically on a chain and on a fork, then on the reference network, and reads the provenance of the rewrites from the model’s history.

2 Setup

```
using BayesianNetworks
```

3 A chain

In the chain $\text{Rain} \rightarrow \text{Sprinkler} \rightarrow \text{Wet}$, observing the sprinkler tells us something about the rain (the sprinkler is turned on less on rainy days); switching it on by hand does not.

```
chain = bayesnet(:Rain => [:no, :yes], :Sprinkler => [:off, :on], :Wet => [:dry, :wet];
          mechanisms = [:Sprinkler => :Rain, :Wet => :Sprinkler])
c = BayesModel(chain)
c = bind_cpt(c, [:Rain => [0.7, 0.3],
                  :Sprinkler => [0.5 0.5; 0.9 0.1], # rows = Rain
                  :Wet => [0.9 0.1; 0.2 0.8]])      # rows = Sprinkler
round.(marginal(c, :Rain).table; digits = 4)
```

```
2-element Vector{Float64}:
 0.7
 0.3
```

Observing the sprinkler on shifts the belief about the rain, because a sprinkler that is on is weak evidence for a dry day:

```
round.(marginal(observe(c, :Sprinkler => :on), :Rain).table; digits = 4)
```

```
2-element Vector{Float64}:
 0.9211
 0.0789
```

Switching the sprinkler on by hand changes nothing upstream: the rewritten network has no arrow into Sprinkler left for the information to travel back along.

```
round.(marginal(do_intervention(c, :Sprinkler => :on), :Rain).table; digits = 4)
```

```
2-element Vector{Float64}:
 0.7
 0.3
```

Downstream, the two agree: Wet depends on Sprinkler only through its own mechanism, which neither operation touches.

```
marginal(observe(c, :Sprinkler ⇒ :on), :Wet).table ≈
  marginal(do_intervention(c, :Sprinkler ⇒ :on), :Wet).table
```

```
true
```

4 A fork

In the fork $\text{Season} \rightarrow \text{Rain}$ and $\text{Season} \rightarrow \text{Sprinkler}$, rain and sprinkler are dependent through the common cause. Observing rain changes the belief about the sprinkler; making it rain does not.

```
fork = bayesnet(:Season ⇒ [:dry, :wet], :Rain ⇒ [:no, :yes], :Sprinkler ⇒ [:off, :on];
  mechanisms = [:Rain ⇒ :Season, :Sprinkler ⇒ :Season])
f = BayesModel(fork)
f = bind_cpt(f, [:Season ⇒ [0.5, 0.5],
  :Rain ⇒ [0.9 0.1; 0.3 0.7], # rows = Season
  :Sprinkler ⇒ [0.2 0.8; 0.8 0.2]]) # rows = Season
(marginal(f, :Sprinkler).table,
  marginal(observe(f, :Rain ⇒ :yes), :Sprinkler).table,
  marginal(do_intervention(f, :Rain ⇒ :yes), :Sprinkler).table)
```

```
([0.5, 0.5], [0.725, 0.275], [0.5, 0.5])
```

5 What the intervention does to the network

`do_intervention` is a local rewrite of the ACSet: the mechanism of the variable and its inputs are removed, and a mechanism with no inputs and a `PointMassRef` is added. Nothing else changes; the graph loses the incoming edges of the variable. The distribution of the rewritten network is therefore the truncated factorisation ([Pearl 2009](#)): the same product of conditional probability tables, with the factor for the intervened variable replaced by a point mass.

```
m = reference_habitat_model()
md = do_intervention(m, :Vegetation ⇒ :dense)
mechanism_names(syntax(md))
```

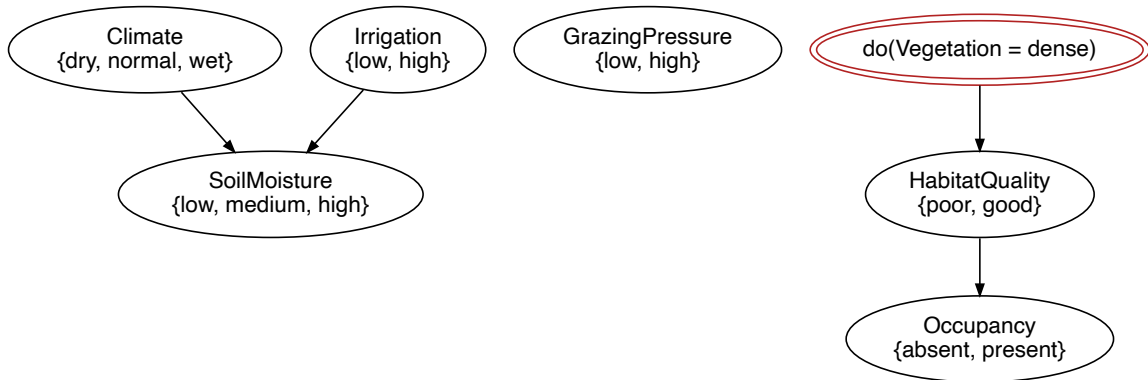
```
7-element Vector{Symbol}:
 :SoilMoisture_mechanism
 :GrazingPressure_mechanism
 :HabitatQuality_mechanism
 :Occupancy_mechanism
 :Climate_mechanism
 :Irrigation_mechanism
 Symbol("do[Vegetation=dense]")
```

```
Symbol[variable_name(syntax(md), p) for p in parents(syntax(md), :Vegetation)]
```

```
Symbol[]
```

Vegetation has no parents left, and the drawing shows the two arrows that used to enter it are gone:

```
to_graphviz(md)
```



The point mass is materialised on demand as a kernel, so evaluation works as before:

```
kernel(md, :Vegetation).table
```

```
3-element Vector{Float64}:
 0.0
 0.0
 1.0
```

Upstream of Vegetation, observation moves SoilMoisture and intervention does not; downstream, both give the same numbers.

```
mo = observe(m, :Vegetation => :dense)
(round.(marginal(m, :SoilMoisture).table; digits = 4),
 round.(marginal(mo, :SoilMoisture).table; digits = 4),
 round.(marginal(md, :SoilMoisture).table; digits = 4))
```

```
([0.288, 0.403, 0.309], [0.0834, 0.35, 0.5666], [0.288, 0.403, 0.309])
```

```
marginal(mo, :Occupancy) ≈ marginal(md, :Occupancy)
```

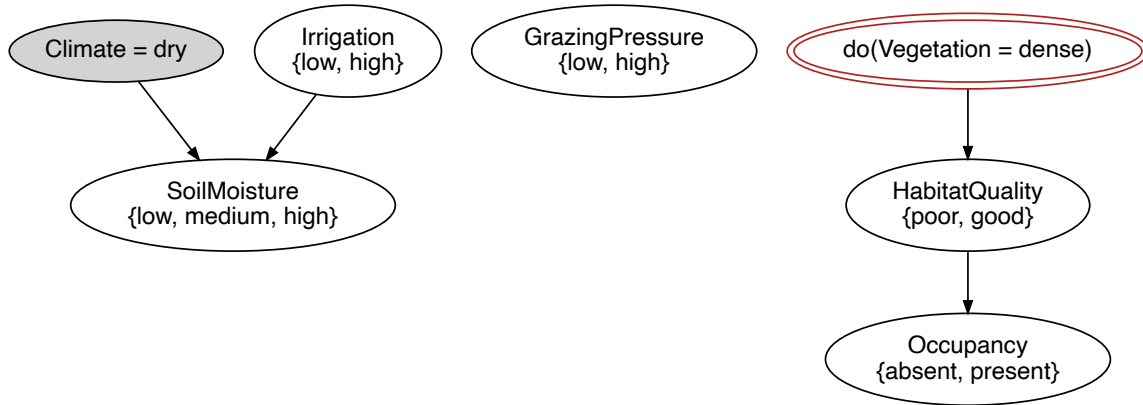
```
true
```

Evidence lives in the wrapper, so a model can carry both: observe on an intervened model conditions the interventional distribution.

```
mdo = observe(md, :Climate => :dry)
evidence(mdo), intervened_variables(mdo)
```

```
(Dict(:Climate => :dry), [:Vegetation])
```

```
to_graphviz(mdo)
```



6 Downstream agreement needs a screening-off condition

The previous example does not establish that observation and intervention always agree downstream. An unobserved common cause can also affect a downstream target through a path that bypasses the manipulated variable:

```
confounded = bayesnet(:U => [:no, :yes], :X => [:no, :yes], :Y => [:no, :yes];
                  mechanisms = [:X => :U, :Y => (:U, :X)])
response = zeros(2, 2, 2)
for u in 1:2, x in 1:2
    response[u, x, u == 2 && x == 2 ? 2 : 1] = 1.0
end
cm = bind_cpt(BayesModel(confounded),
              [:U => [0.5, 0.5], :X => [1.0 0.0; 0.0 1.0], :Y => response])
(observed = marginal(observe(cm, :X => :yes), :Y).table,
intervened = marginal(do_intervention(cm, :X => :yes), :Y).table)
```

```
(observed = [0.0, 1.0], intervened = [0.5, 0.5])
```

Observing $X=\text{yes}$ also reveals $U=\text{yes}$, giving $P(Y=\text{yes})=1$. Setting $X=\text{yes}$ does not reveal or change U , giving $P(Y=\text{yes})=0.5$. The edge $X \rightarrow Y$ is present in both models: being downstream does not remove confounding.

7 Soft interventions

A soft intervention replaces a mechanism by another one, with a chosen set of parents and a kernel supplied by the caller. Here `GrazingPressure` is made to depend on `Climate` (grazing is reduced in dry years):

```
k = cpt([axis(syntax(m), :Climate)], axis(syntax(m), :GrazingPressure),
        [0.8 0.2; 0.5 0.5; 0.3 0.7])      # rows = Climate
ms = soft_intervention(m, :GrazingPressure => k)
variable_name.(Ref(syntax(ms)), parents(syntax(ms), :GrazingPressure))
```

```
1-element Vector{Symbol}:
 :Climate
```

```
round.(marginal(m, :Vegetation).table; digits = 4), round.(marginal(ms,
:Vegetation).table; digits = 4)
```

```
([0.3611, 0.3798, 0.259], [0.3505, 0.3902, 0.2593])
```

The result is validated: a soft intervention that would create a cycle is rejected.

```
try
    soft_intervention(m, :Climate => NamedRef("k"); parents = [:Occupancy])
catch e
    sprint(showerror, e)
end
```

```
"CyclicBayesNetError: the variable graph is not acyclic; variables involved in or
downstream of a cycle: Climate, SoilMoisture, Vegetation, HabitatQuality, Occupancy
(ids 1, 3, 5, 6, 7)"
```

8 Provenance

Every rewrite is recorded in the model's history as a `ModelEvent` with the mechanism that was removed and the one that was added, so an intervened model explains itself.

```
m2 = soft_intervention(md, :GrazingPressure => k; note = "reduced grazing in dry
years")
for e in history(m2)
    println(e.kind, " on ", e.target)
    println("  removed: ", e.removed)
    println("  added:   ", e.added)
    isempty(e.note) || println("  note:    ", e.note)
end
```

```

hard on Vegetation
  removed: MechanismRecord(:Vegetation_mechanism, NamedRef("Vegetation_mechanism"),
    [:SoilMoisture, :GrazingPressure])
  added: MechanismRecord(Symbol("do[Vegetation=dense]"), PointMassRef(:dense),
    Symbol[]))
soft on GrazingPressure
  removed: MechanismRecord(:GrazingPressure_mechanism,
    NamedRef("GrazingPressure_mechanism"), Symbol[])
  added: MechanismRecord(Symbol("soft[GrazingPressure]"),
    NamedRef("soft[GrazingPressure]"), [:Climate])
  note: reduced grazing in dry years

```

```

intervened_variables(m2), is_intervened(m2, :Vegetation), is_intervened(m2,
:Occupancy)

```

```
([:Vegetation, :GrazingPressure], true, false)
```

The events are part of the JSON written by `json_model`, so the provenance travels with the model (see the serialisation vignette).

9 Summary

Observation is a fact about the semantics and leaves the syntax alone; intervention is an edit of the syntax, deleting one mechanism and putting a constant in its place, so their downstream answers agree only when the relevant screening-off/no-confounding conditions hold, and can otherwise differ there as well as upstream. Soft interventions replace a mechanism by another one rather than by a constant, and every rewrite is recorded in the model’s history, so an intervened model can say what was done to it. The next vignette, [Serialisation, provenance and CatColab export](#), writes these models to files and reads them back. Taking networks apart instead of editing them in place is the subject of `CategoricalBayesianNetworks.jl`’s open-networks vignette.

10 References

- Jacobs, Bart, Aleks Kissinger, and Fabio Zanasi. 2019. “Causal Inference by String Diagram Surgery.” *Foundations of Software Science and Computation Structures (FoSSaCS 2019)*, Lecture notes in computer science, vol. 11425: 313–29. https://doi.org/10.1007/978-3-030-17127-8_18.
- Lorenz, Robin, and Sean Tull. 2023. *Causal Models in String Diagrams*. <https://arxiv.org/abs/2304.07638>.
- Pearl, Judea. 2009. *Causality: Models, Reasoning, and Inference*. 2nd ed. Cambridge University Press. <https://doi.org/10.1017/CBO9780511803161>.